materials for weapons use may influence the choice of reactor type. Many dozens of varying reactor concepts have been formulated, so we must limit the discussion in this chapter to a summary of the main variables, and the most common research and power reactors. Fast reactors and some other designs are discussed in Chapter 20.

We have already mentioned three basic principles for reactor design: the neutron energy (thermal or fast reactors), the core configuration (homogeneous or heterogeneous aggregation of fuel and coolant), and the fuel utilization (burner, converter or breeder). In the homogeneous reactor the core can be molten metal, molten salt, an aqueous or an organic solution. In heterogeneous reactors the fuel is mostly rods filled with metal oxide. The fuel material can be almost any combination of fissile and fertile atoms in a mixture or separated as in the core (fissile) and blanket (fertile) concept (Ch. 20). The choice of moderator is great: H_2O, D_2O, Be, graphite, or organic liquid. There is even more choice in the coolant which can be molten metal, molten salt, liquid H_2O, D_2O, or organic liquid, as well as gaseous CO_2, helium or steam.

Some of the more important combinations are summarized in Tables 19.1 and 20.1.

19.12. RESEARCH AND TEST REACTORS

According to the World Nuclear Industry Handbook 2012, 264 research and test reactors were in operation throughout the world at the end of 2011, but some have probably been shut down since then. While the smaller research reactors may have an operating power $\leq 1\ kW_{th}$, and do not need forced cooling, the larger test reactors operate at $\leq 50\ MW_{th}$. Many of these reactors have facilities for commercial radionuclide production.

Research reactors are used in nuclear physics, in nuclear, analytical and structural chemistry, in radiobiology, in medicine, etc. They are usually easy to operate, inherently safe, and of moderate cost. Many of them are of the pool type; the reactor core is located in the center of a stainless steel clad concrete vessel, 6−8 m deep and 2−5 m wide, containing purified water. The water provides the main radiation shielding, moderation, and cooling. The concrete walls are about 2 m thick when the reactor is located above ground; otherwise much thinner walls can be used. The main radiation protection demand comes from the reaction

$$^{16}O(n,p)^{16}N \xrightarrow[7.13\ s]{\beta\gamma} {}^{16}O \tag{19.39}$$

since the γ's emitted are very energetic (6.1 MeV). At power levels $>100\ kW_{th}$ forced cooling may be required. The fuel contains usually highly enriched uranium (20 to 90% ^{235}U) as rods or plates; the amount of ^{235}U is ≤ 3 kg.

Some reactors are designed to produce very high neutron fluxes, either for testing materials (especially for fast reactors) or for isotope production. The high flux reactor at Grenoble, France, has fuel plates of highly enriched (93%) UAl_3 canned in aluminum.

The construction is a swimming pool tank type. Using 8.6 kg ^{235}U and 15 m^3 heavy water as moderator, it provides a maximum thermal neutron flux of 1.5×10^{19} m^{-2} s^{-1} at 60 MW_{th}; 10 fold higher fluxes can be achieved during shorter periods. At Oak Ridge, the High Flux Isotope Reactor (HFIR) has a flux of 3×10^{19} m^{-2} s^{-1}, the highest continuous thermal neutron flux so far reported. Some research reactors are built to test new reactor designs which may be candidates for future power reactors. The successful designs lead to prototype power reactors; some of these are described in Chapter 20 together with full-scale power reactors. Other designs never survive the first stage, although the design may be very interesting.

19.13. THERMAL POWER REACTORS

We illustrate the general principles of thermal reactors by a short description of the two most important power reactor types; the pressurized water reactor (PWR) and the boiling water reactor (BWR). They are further discussed in Chapter 20.

19.13.1. The pressurized water reactor

As the name implies a pressurized water reactor (PWR) is cooled by hot high pressure water, H_2O, which also acts as moderator. The main components of a pressurized light water moderated and cooled reactor (PWR) station have been described in §19.1 and Figure 19.2. The Ringhals station, whose first reactor started operation in 1975, is located in Sweden near the sea and seawater is used for cooling; power stations located near large rivers or at the sea shore usually do not need cooling towers.

The important features of the PWR core are shown in Figure 19.12, which, although taken from three different reactors, represents the typical Westinghouse design. A typical core contains ~40 000 fuel rods (a) in 193 assemblies, each with space for 208 fuel pins (b) and (c). The fuel is 88 t of UO_2 with an enrichment of 2.17% (inner region) to 2.67% (outer region) in the initial fuel loading. The reactor vessel (e) is made of stainless steel clad low alloy steel, 13 m high and 4.4 m in diameter, with a wall thickness of 22 cm; (c) shows the distribution of control rods entering from the top of the vessel. The 1060 control rods are made of a silver alloy containing 15 wt% In and 5 wt% Cd; both these elements have high thermal neutron capture cross sections. The main reactivity control is by boric acid, which is fed to the coolant through a special injection system. The boric acid circulates in the primary coolant loop, and acts as a neutron absorber. At the start of a fresh core its concentration is ~1500 ppm (~0.025 M), but it is successively reduced to zero at time of fuel replacement. The concentration is adjusted in a side loop, containing either an anion exchange or an evaporator system. The ion exchange system is so designed that H_3BO_3 is fixed to the ion exchanger (the water is deborated) at low temperature (~30°C) while it is eluted at higher temperature (~80°C).